

C:\Users> Afroz > Desktop > EXP8 > calculator.y

```
1  %{
2  #include <stdio.h>
3  #include <stdlib.h>
4
5  int yylex(void);
6  void yyerror(const char *s);
7  %}
8
9  %token NUMBER
10
11  %left '+' '-'
12  %left '*' '/'
13  %right UMINUS
14
15  %%
16
17  input:
18  | /* empty */
19  | input line
20  ;
21
22  line:
23  | '\n'
24  | expr '\n' { printf("Result = %d\n", $1); }
25  ;
26
27  expr:
28  | NUMBER
29  | expr '+' expr { $$ = $1 + $3; }
30  | expr '-' expr { $$ = $1 - $3; }
31  | expr '*' expr { $$ = $1 * $3; }
32  | expr '/' expr { $$ = $1 / $3; }
33  | '(' expr ')' { $$ = $2; }
34  | '-' expr %prec UMINUS { $$ = -$2; }
35  ;
36
37  %%
38
39  void yyerror(const char *s)
40  {
41  | printf("Error: %s\n", s);
42  }
43
44  int main()
45  {
46  | printf("Enter an arithmetic expression:\n");
47  | yyparse();
48  | return 0;
49  }
50
```

```
lexer.l
C: > Users > Afroz > Desktop > EXP8 > lexer.l
1  %{
2  #include "calculator.tab.h"
3  #include <stdlib.h>
4  %}
5
6  %%
7
8  [0-9]+      { yylval = atoi(yytext); return NUMBER; }
9  [ \t]      ;
10 \n         { return '\n'; }
11 .          { return yytext[0]; }
12
13 %%
14
15 int yywrap()
16 {
17     return 1;
18 }
19
```

```
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP8
$ ./calculator
Enter an arithmetic expression:
10+5*2
Result = 20
```